

TIANSHUI DAEI, China

WMO: 570140

Lat: 34.5614N

Lon: 105.8739E

Elev: 3560

StdP: 12.90

Time Zone: 8.00 (E08)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	17.1	0.3	6.4	26.9	3.6	7.5	28.2	15.3	29.2	14.0	30.3	3.0	290	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	90.6	68.3	88.2	67.9	85.9	67.3	72.7	83.9	71.5	82.8	70.4	80.8	7.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	124.4	77.5	68.3	118.9	76.5	67.2	114.2	75.9	39.1	85.2	37.9	83.0	36.8	80.8	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.2	12.2	10.7	DB	8.3	95.2	3.6	2.9	5.7	97.3	3.6	98.9	1.6	100.6	-1.0	102.7
			WB	7.6	73.7	3.6	1.6	5.0	74.9	2.9	75.9	0.9	76.8	-1.7	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	30.3	37.0	48.2	58.3	64.6	71.3	75.6	74.3	64.5	54.7	43.0	32.0	
	DBStd	16.54	5.60	6.33	7.41	6.33	5.17	3.96	4.15	5.16	4.99	5.49	6.05	4.54	
	HDD50	1908	611	367	124	11	0	0	0	0	0	20	216	559	
	HDD65	4741	1076	785	521	213	71	4	0	2	66	320	659	1024	
	CDD50	3580	0	2	68	259	452	640	793	755	437	167	7	0	
	CDD65	938	0	0	0	11	59	194	328	292	52	2	0	0	
	CDH74	7740	0	0	15	198	638	1547	2752	2324	248	18	0	0	
	CDH80	2557	0	0	2	24	135	498	1040	813	43	2	0	0	
Wind	WSAvg	4.4	4.1	4.6	5.0	4.9	4.8	4.5	4.8	4.9	4.2	3.7	3.6	3.6	
Precipitation	PrecAvg	21.2	0.2	0.2	0.7	1.7	2.2	2.6	3.7	3.5	3.6	1.9	0.6	0.2	
	PrecMax	30.4	0.4	0.7	1.7	3.2	4.8	6.9	7.7	8.6	7.1	4.5	1.6	0.7	
	PrecMin	12.8	0.0	0.0	0.0	0.4	0.4	0.6	1.0	0.6	1.3	0.5	0.1	0.0	
	PrecStd	4.8	0.1	0.2	0.4	0.8	1.1	1.4	1.6	2.1	1.8	1.0	0.4	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	62.6	75.5	83.3	87.3	91.8	94.2	93.9	85.2	76.1	65.1	51.5	
		MCWB	39.6	45.0	52.5	58.2	61.2	64.5	70.2	70.7	64.7	60.0	51.3	39.4	
	2%	DB	48.1	56.7	70.5	79.1	83.6	88.6	91.0	89.8	79.8	71.0	60.8	47.7	
		MCWB	36.8	42.8	50.6	56.3	60.8	64.4	69.3	69.9	64.4	58.1	49.3	37.4	
	5%	DB	43.9	52.7	66.3	75.9	80.7	85.7	88.6	87.3	76.5	67.6	56.3	44.6	
		MCWB	34.1	40.6	48.7	55.4	60.0	64.4	68.8	69.6	63.4	56.4	46.3	35.8	
	10%	DB	40.3	48.3	62.0	72.0	77.7	82.6	86.0	84.8	73.6	64.3	53.1	41.7	
		MCWB	32.2	38.3	46.6	53.8	59.3	64.4	68.7	68.8	62.3	54.8	45.2	33.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.9	47.4	55.7	62.2	65.9	71.0	74.8	74.3	71.5	62.1	52.8	41.2
			MCDB	51.1	60.5	68.8	79.5	78.8	80.3	85.5	84.2	76.1	72.5	63.3	49.4
2%		WB	37.4	43.7	52.2	59.2	64.1	69.2	72.9	72.8	67.4	59.9	50.0	38.6	
		MCDB	46.5	54.5	67.6	73.8	76.6	78.7	83.9	84.6	74.0	67.3	57.7	45.8	
5%		WB	34.9	41.3	49.8	57.3	62.8	67.8	71.7	71.6	65.3	58.4	48.4	36.6	
		MCDB	42.6	51.1	63.6	70.7	75.3	78.8	83.0	83.7	71.7	65.2	54.3	43.6	
10%		WB	32.6	39.0	47.6	55.3	61.3	66.3	70.6	70.5	63.9	56.7	46.5	34.4	
		MCDB	39.7	46.8	59.5	68.3	72.8	77.5	81.2	81.6	70.3	62.6	51.7	40.6	
Mean Daily Temperature Range	5% DB	MDBR	18.4	18.1	20.6	22.6	21.3	19.6	17.7	17.0	14.7	16.2	17.9	19.1	
		MCDBR	25.9	27.3	29.5	30.0	29.2	28.2	24.2	21.5	21.6	22.7	24.9	24.2	
		MCWBR	16.0	15.2	13.0	11.2	10.0	8.2	6.7	5.8	8.0	10.0	13.6	15.0	
	5% WB	MCDBR	24.2	25.9	26.9	24.1	21.6	19.3	16.8	16.6	14.0	18.4	21.1	22.1	
		MCWBR	15.4	15.0	12.8	10.3	8.6	6.9	5.7	5.2	6.0	9.2	12.1	14.2	
Clear-Sky Solar Irradiance	taub	0.481	0.552	0.599	0.599	0.575	0.559	0.561	0.602	0.562	0.542	0.484	0.440		
	taud	1.775	1.660	1.609	1.648	1.766	1.858	1.907	1.782	1.831	1.803	1.841	1.889		
	Ebn at Noon	213	212	217	227	235	238	237	223	224	213	211	217		
	Edh at Noon	55	69	79	79	71	65	62	69	62	59	51	46		
All-Sky Solar Radiation	RadAvg	816	969	1292	1589	1683	1679	1593	1430	1048	910	813	767		
	RadStd	79	142	118	152	144	118	207	175	104	133	122	97		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-160	+126	N/A

Nomenclature: See separate page